KUYFI

BLACK-BOX SECURITY REPORT

Before the Breach.

Contract ID	CCEUDLQ2HQSW5NXBLXBUNUBIZQK6O5FAGESEQYK6H4GQXY36CZDKCLUZ
Network	TESTNET
Report ID	kyf-20261025T163802Z-6aedf75e
Generated At	2026-10-25T16:38:02.475Z
Tool Version	Kuyfi 0.1.0
Schema Version	1.0.0

This report is generated by automated black-box fuzzing against a deployed Soroban smart contract. It is not a manual audit and does not constitute a formal security certification.

2. Executive Summary

Functions detected	11
UDTs detected	3
Vectors executed	92
Broadcast transactions	16
Transactions with hash	16
Total findings	0

Findings by Severity

Critical	0
High	0
Medium	0
Low	0
Info	0

3. Attack Surface

Target

Contract ID	CCEUDLQ2HQSW5NXBLXBUNUBIZQK6O5FAGESEQYK6H4GQXY36CZDKCLUZ
Network	TESTNET
Bytecode size	10592 bytes

Functions Detected (11)

- `close(chest: Address) -> returns value`
- `set_fees(creation_fee: I128, daily_fee_bps: U32, max_duration_days: U32) -> returns value`
- `get_config() -> returns value`
- `chest_count() -> returns value`
- `list_chests(start: U32, limit: U32) -> returns value`
- `accept_admin() -> returns value`
- `create_chest(creator: Address, question: String, fingerprint: BytesN<32>, amount_per_slot: I128, slots: U32, duration_days: U32) -> returns value`
- `set_treasury(treasury: Address) -> void`
- `__constructor(admin: Address, treasury: Address, chest_wasm_hash: BytesN<32>, verifier: Address, token: Address, creation_fee: I128, daily_fee_bps: U32, max_duration_days: U32) -> void`
- `propose_admin(new_admin: Address) -> void`
- `set_chest_wasm(chest_wasm_hash: BytesN<32>) -> void`

UDTs Detected (3)

STRUCT Config

- `admin: Address`
- `chest_wasm_hash: BytesN<32>`
- `creation_fee: I128`
- `daily_fee_bps: U32`
- `max_duration_days: U32`
- `token: Address`
- `treasury: Address`
- `verifier: Address`

UNION DataKey

- `Config`
- `Count`
- `ChestAt(U32)`
- `PendingAdmin`

STRUCT ChestInit

- `amount_per_slot: I128`
- `creator: Address`
- `daily_fee_bps: U32`
- `duration_days: U32`
- `fee_reserve: I128`
- `fingerprint: BytesN<32>`
- `question: String`
- `slots: U32`
- `token: Address`
- `treasury: Address`
- `verifier: Address`

4. Chaos Execution

Kuyfi performs black-box fuzzing against the deployed Soroban contract using its on-chain contract specification.

Vectors executed	92
Broadcast transactions	16
Transactions with hash	16
Verification transactions	2

5. Findings

NO REPORTABLE FINDINGS

No reportable security findings were produced by this audit run. This does not constitute a formal proof that the contract is vulnerability-free.

6. Verification

Real Testnet transactions broadcast during this audit run, provided so every claim above can be independently verified on-chain.

list_chests — start::ZERO

Transaction hash	e915b7e8cfdff4f4685e02027d5e6a11fb2bfda29112ac3162618e9ff9095600
Ledger	5384523

Explorer URL
https://stellar.expert/explorer/testnet/tx/e915b7e8cfdff4f4685e02027d5e6a11fb2bfda29112ac3162618e9ff9095600

set_fees — max_duration_days::ONE

Transaction hash	c3ca1b8e3998dc7a6429b320b99e1478cc69e2cd89f153cd4faebc25b44731d0
Ledger	5384520

Explorer URL
https://stellar.expert/explorer/testnet/tx/c3ca1b8e3998dc7a6429b320b99e1478cc69e2cd89f153cd4faebc25b44731d0